Question	Answer	Marks	Partial Marks
2(c)	ruled line at $y = -1$	**B1**	
	0.3 to 0.32	**B1**	
3(a)(i)	169	**2**	**M1** for g(13) or $(1+4x)^2$ or better
3(a)(ii)	$1+4x^2$ final answer	**1**	
3(a)(iii)	x	**1**	
3(b)	3.5 or $\frac{7}{2}$	**2**	**M1** for $1+4x=15$
4(a)(i)	40.9 or 40.91…	**3**	**M2** for [sin ABC=] $\frac{29.5\sin 51.6}{35.3}$ oe or for [cos ABC =] $\frac{35.3^2+45^2-29.5^2}{2\times 35.3\times 45}$ or **M1** for $\frac{29.5}{\sin ABC}=\frac{35.3}{\sin 51.6}$ oe or for correct implicit cosine rule
4(a)(ii)	520 or 520.0 to 520.2…	**2**	**FT** *their* (a)(i) if used provided working shown **M1** for $0.5\times 29.5\times 45\times \sin 51.6$ oe or for $0.5\times 35.3\times 45\times \sin(\textit{their}(\text{a})(\text{i}))$ or for 0.5 × 35.3 × 29.5sin (180–51.6–*their*(a)(i))
4(b)(i)	41.2 or 41.21 to 41.23	**4**	**M1** for $SQ = 2\times 32\times \sin\left(\frac{1}{2}\times 56\right)$ oe or $\sqrt{32^2+32^2-2\times 32\times 32\times \cos 56}$ oe or $\frac{32\sin 56}{\sin((180-56)\div 2)}$ oe **M2** for $SR^2 = 47^2+\left(\textit{their } SQ^2\right)-2\times 47\times \textit{their } SQ\times \cos 60$ or **M1** for implicit form
4(b)(ii)	28.3 or 28.25 to 28.29…	**3**	**M2** for 32 × sin62 oe or **M1** for recognition that line from P is perpendicular to SQ
5(a)	121 or 120.8… or 120 $\frac{5}{6}$	**4**	**M1** for midpoints soi **M1** for use of $\sum fx$ with x in correct interval including both boundaries but **not** if x is 50, 50, 100 and 300 **M1** (dep on 2nd M1) for $\sum fx \div 120$

Question	Answer	Marks	Partial Marks
5(b)	12.4 5 1.4	**3**	**B1** for each If 0 scored **SC1** for fd's [0.86,] 0.62, 0.25 and 0.07 oe
5(c)	43 74 99 120	**2**	**B1** for 2 or 3 correct
5(d)	Correct diagram	**3**	**B1** for correct horizontal placement for 4 plots **B1FT** for correct vertical placement for 4 plots **B1FT dep on at least B1** for reasonable increasing curve or polygon through *their* 4 points If 0 scored **SC1 FT** for 3 out of 4 points correctly plotted
5(e)(i)	Strict FT *their* **median** reading	**1**	
5(e)(ii)	Strict FT *their* **UQ** reading	**1**	
5(e)(iii)	Strict FT *their* reading at **40th percentile**	**2**	**B1** for 48 written or mark at cf = 48 on graph
5(e)(iv)	Strict FT *their* reading at 400 – *their* reading at 250	**2**	**B1** for either correct reading at 250 or 400
6(a)	15	**2**	**M1** for $\frac{360}{180-156}$ or for $\frac{180(n-2)}{n}=156$ oe
6(b)	38	**2**	**B1** for $AOB = 76$
6(c)	68	**2**	**B1** for $RSP = 68$ or $RQP = 112$